

Enea Monzio Compagnoni

Final-year PhD in Computer Science, University of Basel · Zürich, Switzerland
Intern at Google – Intrinsic AI Robotics, Model Distillation, Quantization & Optimization

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Summary

Research-focused final-year PhD candidate in stochastic optimization and modern deep learning, with expected PhD completion in **October 2026**. My work connects SDE-based theory with practical training behavior for adaptive optimizers, compressed/distributed learning, differential privacy, and large-scale neural network training. I derived **batch-size scaling laws** for learning rate, AdamW momentum parameters (β_1, β_2), and decoupled weight decay to guide hyperparameter transfer. I currently explore **RL optimization** ideas, including PPO/GRPO-style objective modifications and hyperparameter schedules, with a focus on stable and reliable training. I validate theoretical results with large-scale PyTorch experiments and turn them into actionable training insights. I mentor PhD/MSc students, collaborate across institutions, and have published work at **ICML, ICLR, AISTATS, Quantitative Finance, and IJCNN**.

Skills

Programming: **Python** (PyTorch), reproducible ML experimentation

ML/Optimization: Stochastic optimization, Transformers/LLMs, differential privacy, RL

Mathematics: Stochastic calculus, probability, linear algebra, statistics

Experience

Google – Intrinsic AI Robotics

Jul 2026 – Dec 2026

Intern – Zürich, Switzerland

- Focus: Model Distillation, Quantization & Optimization

University of Basel

Oct 2022 – Oct 2026 (expected)

PhD Researcher & Lecturer, Computer Science – Basel, Switzerland

- Developed SDE-based analyses for SAM, AdamW, distributed compressed SGD/SignSGD, and differentially-private optimizers.
- Derived and validated **batch-size scaling rules** for learning rate, AdamW momentum parameters (β_1, β_2), gradient-noise scale, and decoupled weight decay.
- Built reproducible experimental pipelines for deep-learning optimization in PyTorch, including Transformers, CNNs, ResNets, and controlled synthetic settings.
- Worked on privacy-aware optimization and LLM training pipelines under high-privacy regimes; connected theory to practical hyperparameter transfer across privacy levels.

Flexion Robotics AG

May 2025 – Aug 2025

AI Engineer – Robotics Intern (Perception) – Zürich, Switzerland

- Implemented **VLM/VLA**-based agentic frameworks for autonomous robot operations.
- Developed advanced prompting strategies; integrated components for reliable sim-to-real operation.
- Designed experimental setups for **stable, reproducible, and goal-oriented** workflows.
- **Optimized** the agent for practical use; aligned stakeholders; coordinated a successful real-world robotics demonstration.

Yahoo Research

Jul 2023 – Sep 2023

Intern, Scalable Machine Learning – Munich, Germany

- Built a risk-aware controller for ad-budget pacing; improved spend control by **~4%** in A/B tests and delivered production-adjacent Python components.

UBS AG

Jul 2019 – Sep 2022

AI Quantitative Analyst – Zürich, Switzerland

- Designed a Lombard lending risk model for UHNW clients; improved decisioning quality.
- Increased Japanese-market **liquidity-forecast** accuracy by **~3%**; enhanced AML detection with new features and ML pipelines.

Publications

- **E. Monzio Compagnoni**, R. Islamov, F.N. Proske, E. Gorbunov, A. Lucchi, A. Orvieto. *On the Interaction of Batch Noise, Adaptivity, and Compression under (L_0, L_1) -Smoothness: An SDE Approach*. ICML, 2026.
- **E. Monzio Compagnoni**, A. Stanghellini, R. Islamov, A. Lucchi, A. Koloskova. *Adaptive Methods Are Preferable in High Privacy Settings: An SDE Perspective*. ICLR, 2026.
- **E. Monzio Compagnoni**, T. Liu, R. Islamov, F.N. Proske, A. Orvieto, A. Lucchi. *Adaptive Methods through the Lens of SDEs: Theoretical Insights on the Role of Noise*. ICLR, 2025.
- **E. Monzio Compagnoni**, R. Islamov, F.N. Proske, A. Lucchi. *Unbiased and Sign Compression in Distributed Learning: Comparing Noise Resilience via SDEs*. AISTATS (Oral), 2025.
- **E. Monzio Compagnoni**, A. Orvieto, H. Kersting, F.N. Proske, A. Lucchi. *SDEs for Minimax Optimization*. AISTATS, 2024.
- M. Burzoni, A. Doldi, **E. Monzio Compagnoni**. *Risk Sharing with Deep Neural Networks*. Quantitative Finance, 2024.
- **E. Monzio Compagnoni**, L. Biggio, A. Orvieto, F.N. Proske, H. Kersting, A. Lucchi. *An SDE for Modeling SAM: Theory and Insights*. ICML, 2023.
- **E. Monzio Compagnoni**, A. Scampicchio, L. Biggio, A. Orvieto, T. Hofmann, J. Teichmann. *On the Effectiveness of Randomized Signatures as Reservoir for Learning Rough Dynamics*. IJCNN, 2023.

Education

University of Basel

Oct 2022 – Oct 2026 (expected)

PhD in Computer Science

ETH Zürich

Sep 2018 – Apr 2022

MSc in Quantitative Finance (GPA 5.76/6, Summa Cum Laude)

University of Milan

2012 – 2018

MSc & BSc in Mathematics (110/110, Summa Cum Laude)

References

- PhD Supervisor: Prof. Dr. Aurelien Lucchi: aurelien.lucchi@unibas.ch
- External Collaborator: Dr. Antonio Orvieto: antonio@tue.ellis.eu

Languages

Italian, English